

Chinese Input Method Editor Research in Extended Reality

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Input Method Editor

- What is this?
- You will use it if you need to input some non-Latin symbol (中文, 日本語, 조선어)
- We have a mature system in qwerty keyboard, but not in Mixed Reality.

A short message in MR



“早上好”

“Good morning”

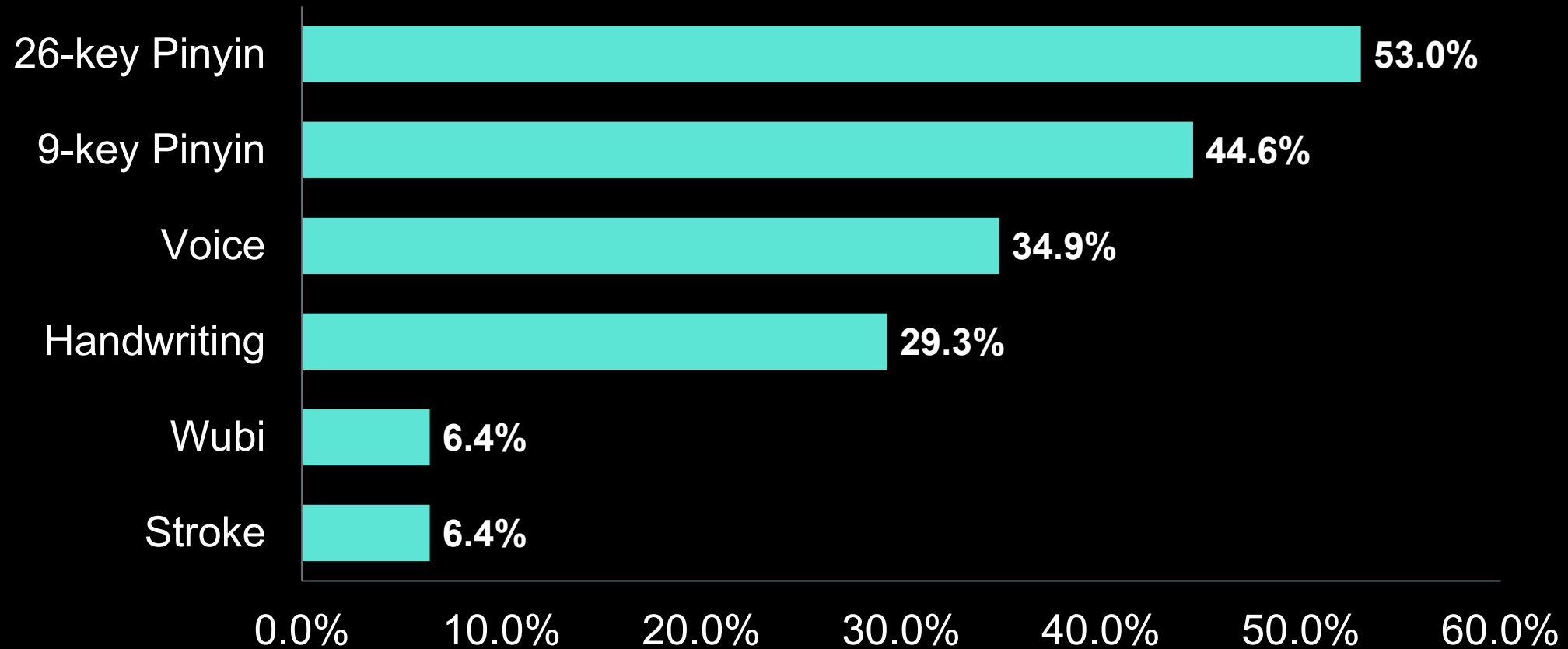
Enter a message.
Choose the intended text.
Recover from a mistake.

Who can see the message?

Challenges – Error happens

Stage	Illustrative example
Input and composition	zaoshanghao
Candidate selection	枣上号/早上好
Commit and correction	Choose 早上号, then correct 号 to 好

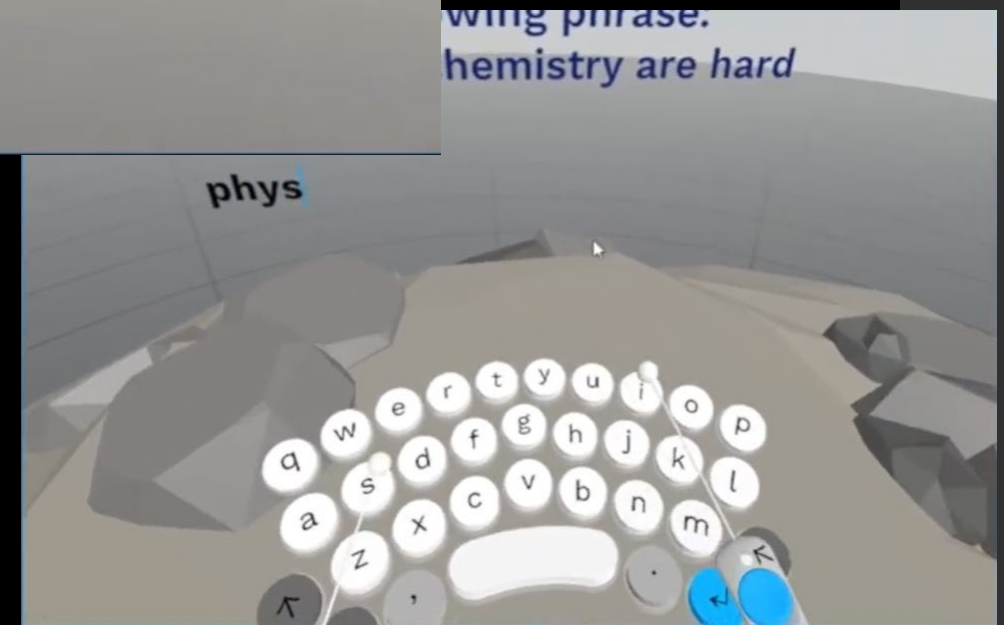
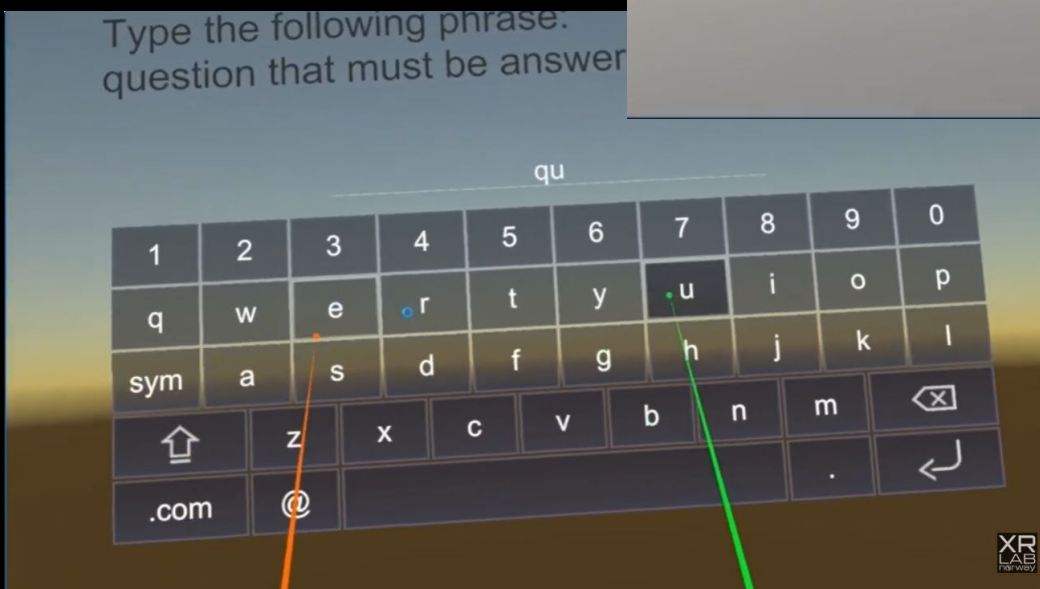
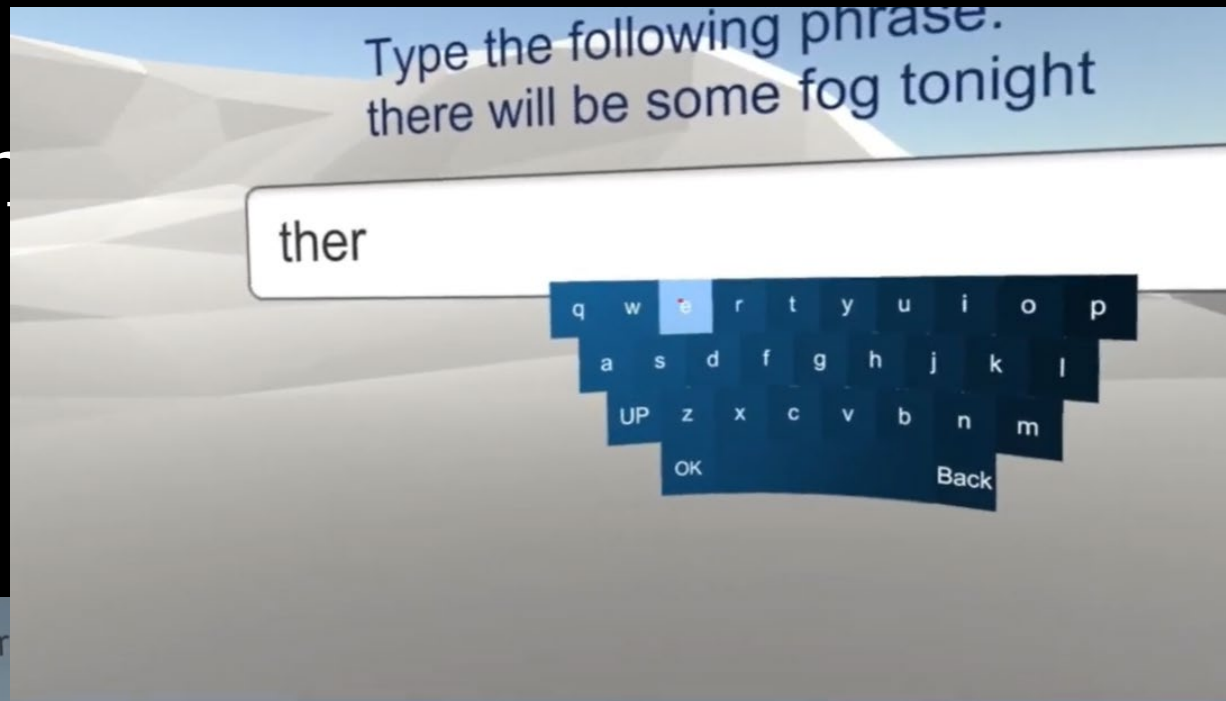
Challenges – Different environment



Source: CTR, second-half 2020 survey, p. 13.

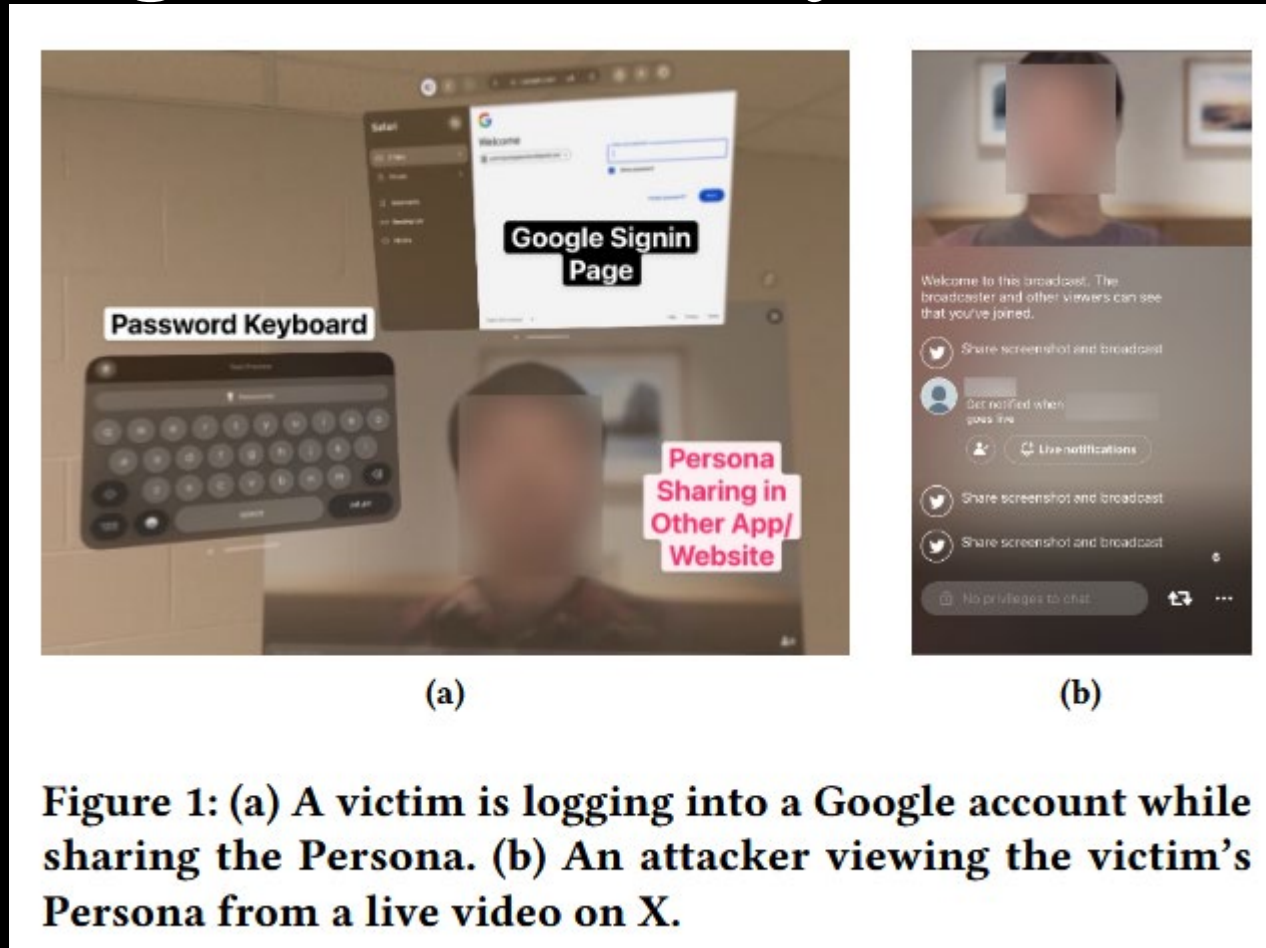
Challenge

ment



Source: **VR text input techniques: Ray-casting, Drum-like keyboard, Head-directed input, Split keyboard**
. <https://www.youtube.com/watch?v=DRqOQ7-Uv6E>

Challenges – Security and Privacy

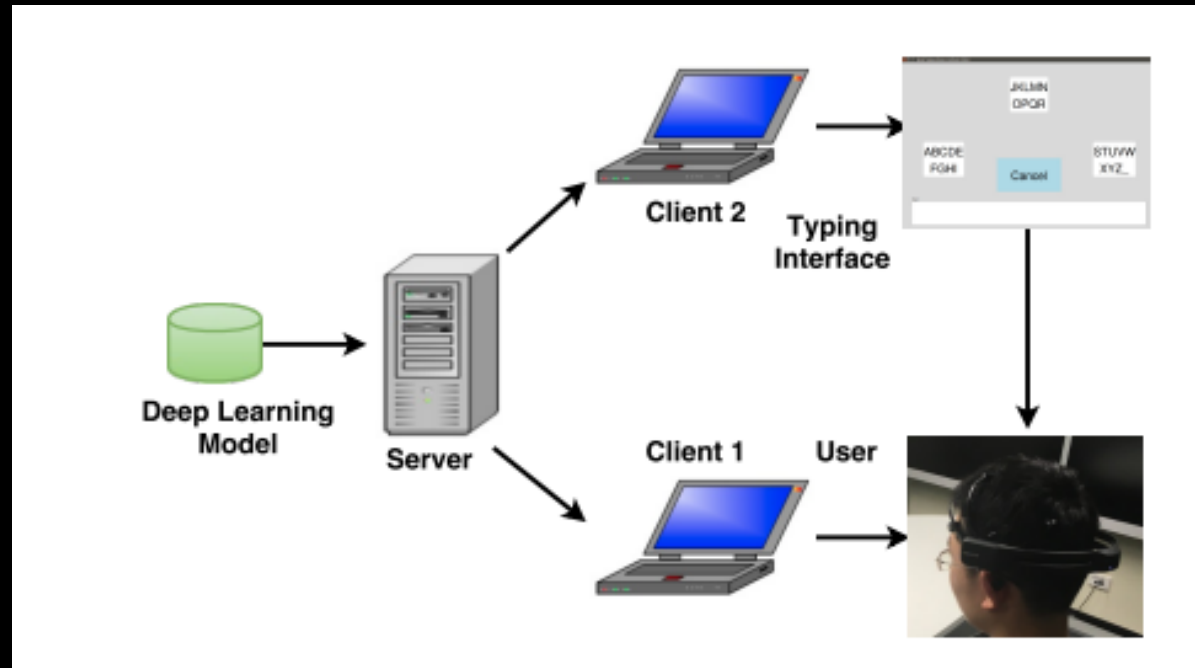


Some Solutions

- Swipe input
- Different keyboard layout
- Alternative input (Voice, EEG to text, physical keyboard, eye input)

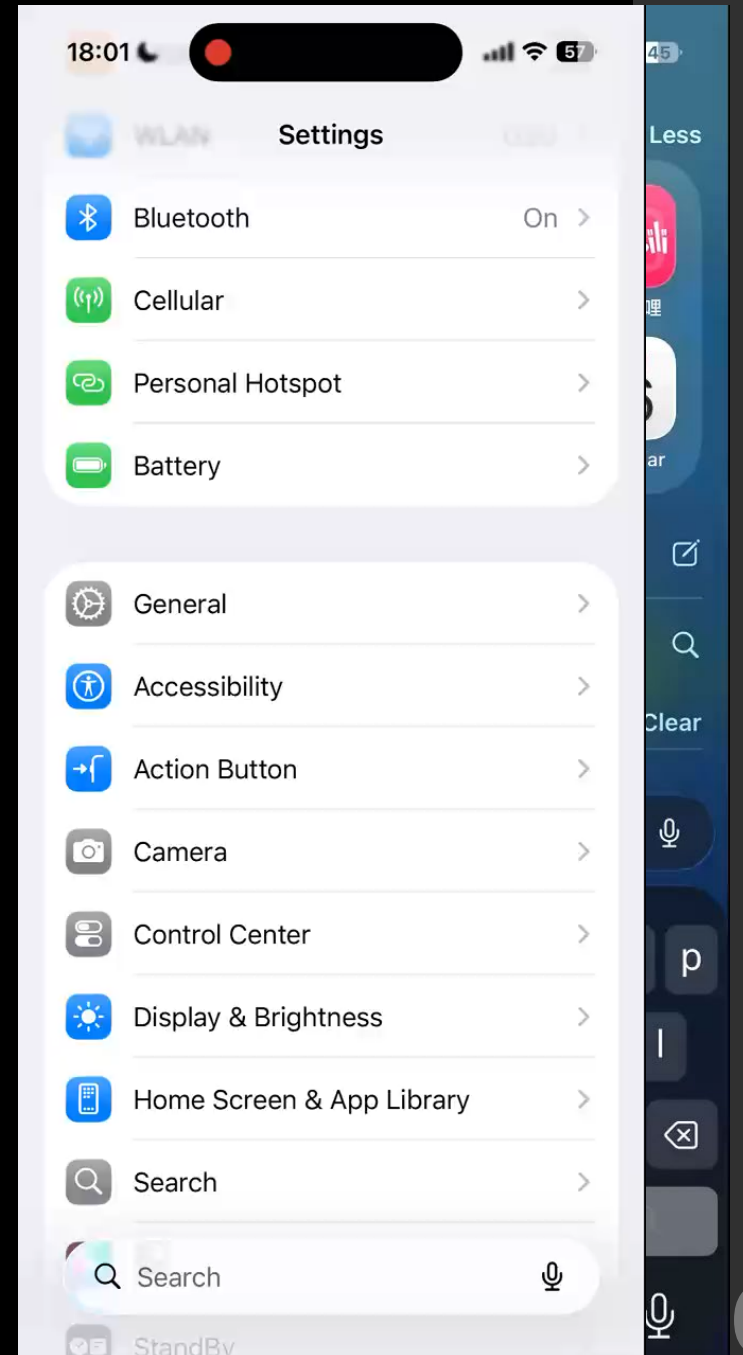
EEG to text

- 5 operation with 95% Accuracy.
- And two complex model (RNN+CNN)



Swipe input

- Slide to type (IOS)
- My favourite input method after I get a smart phone.
- Very fast and inaccuracy input even in English.
- Higher mental pressure.



Swipe input - Question

Interruption

Can a user pause and resume a continuous action?

Correction

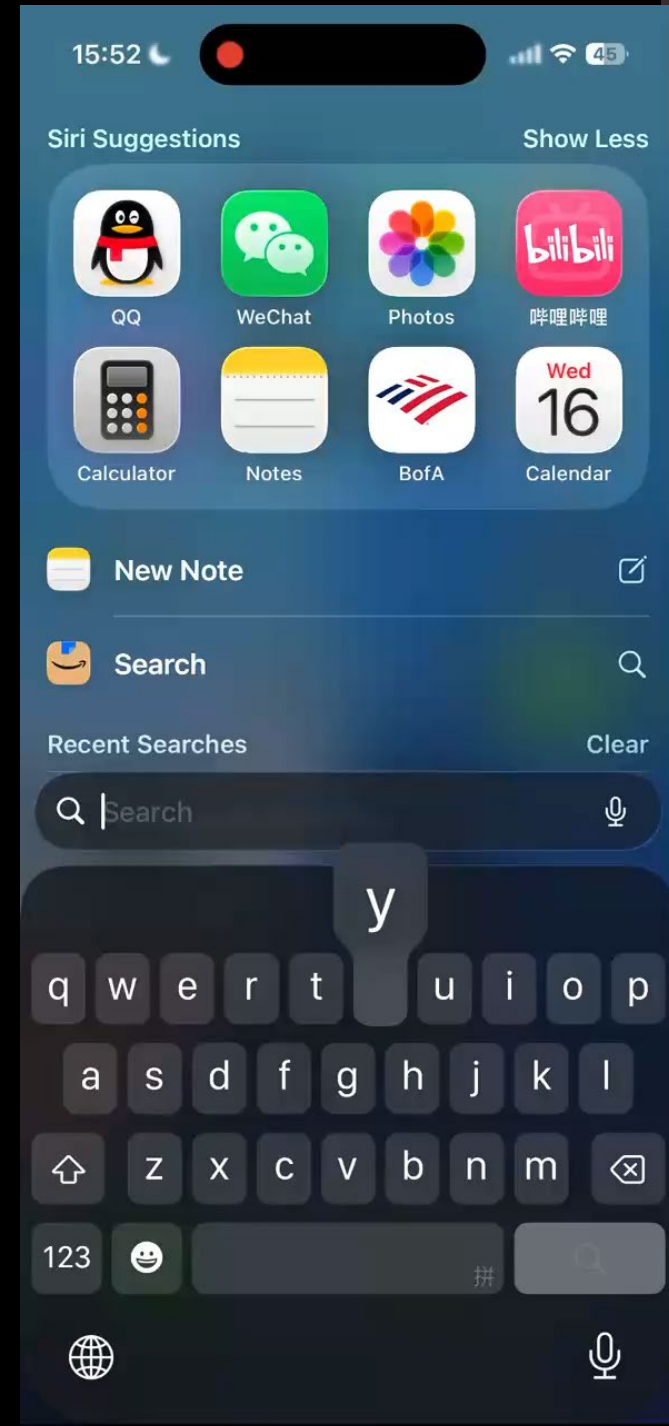
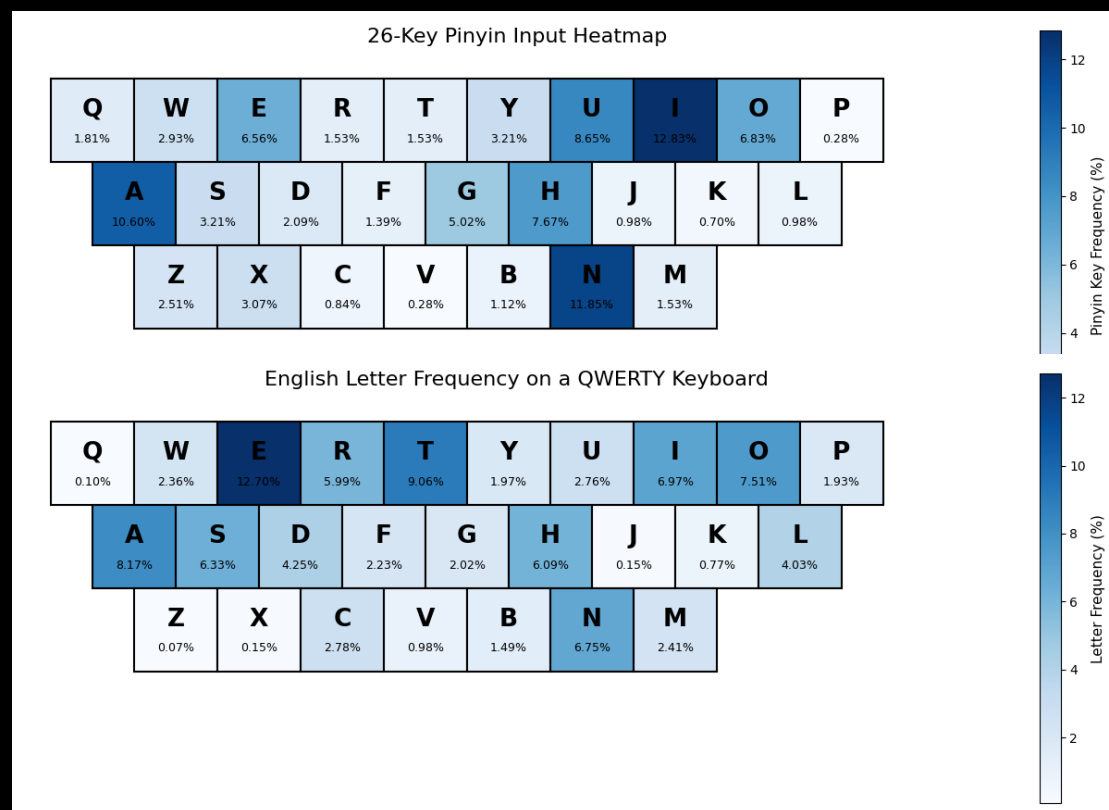
How much work does one inaccurate movement create?

Chinese composition

How do ambiguous input and candidate selection interact with a gesture path?

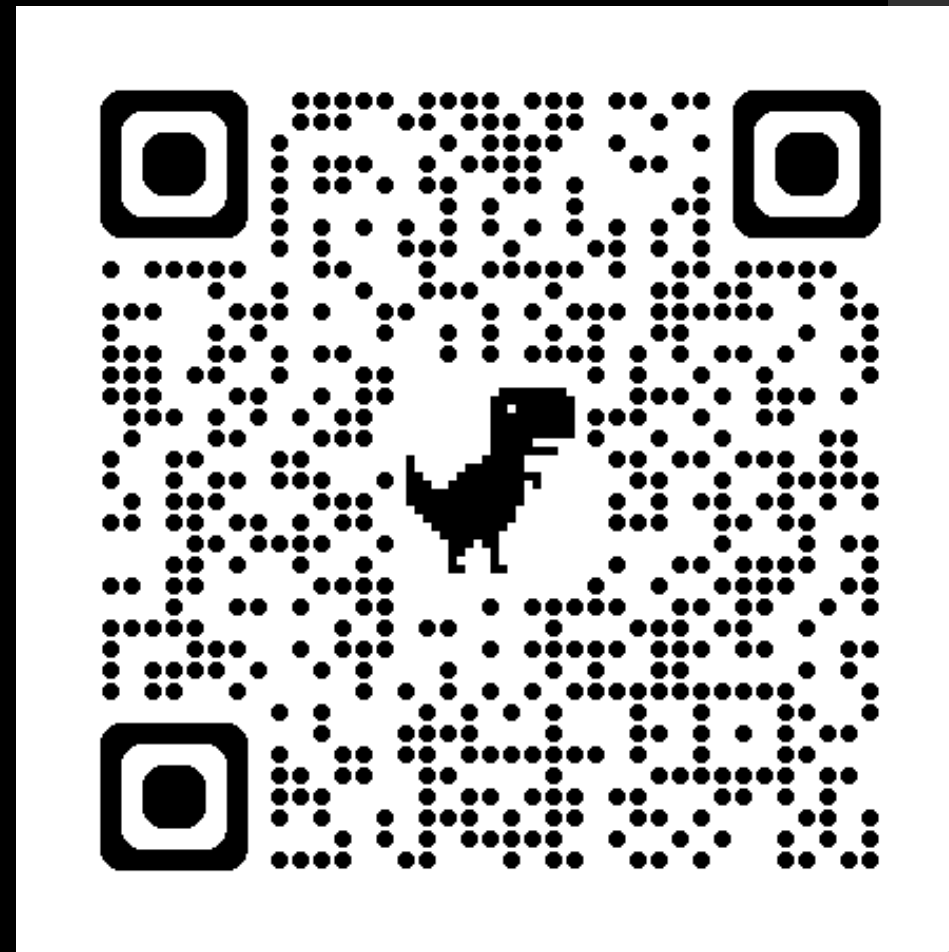
Swipe input - Question

一天 (yī'tian) , 一团(yī'tuan), 一条(yī'tiao)



Chinese Input Method Editor

- In Apple's IOS, we have 8 default IME:
- Pinyin – QWERTY, 10 Key, AZERTY
- Shuangpin (Two-initial Pinyin)
- Handwriting
- Stroke
- Wubi
- Zhuyin – Standard, Dynamic
- Sucheng – QWERTY, Standard
- Cangjie – QWERTY, Standard



5-min IME questionnaire

Current progress

Available now

Background material

An early stroke-input mockup

Questions about interaction

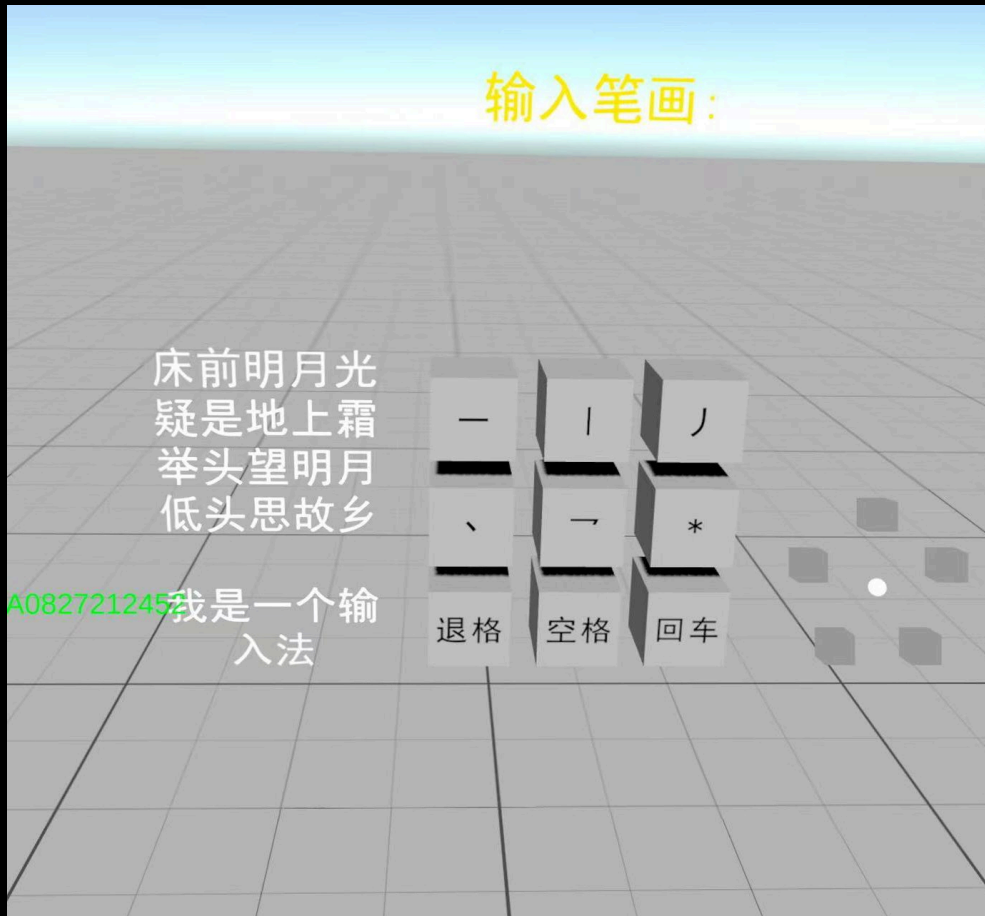
Still to establish

End-to-end IME behavior

Usability evidence

Identity or security implications

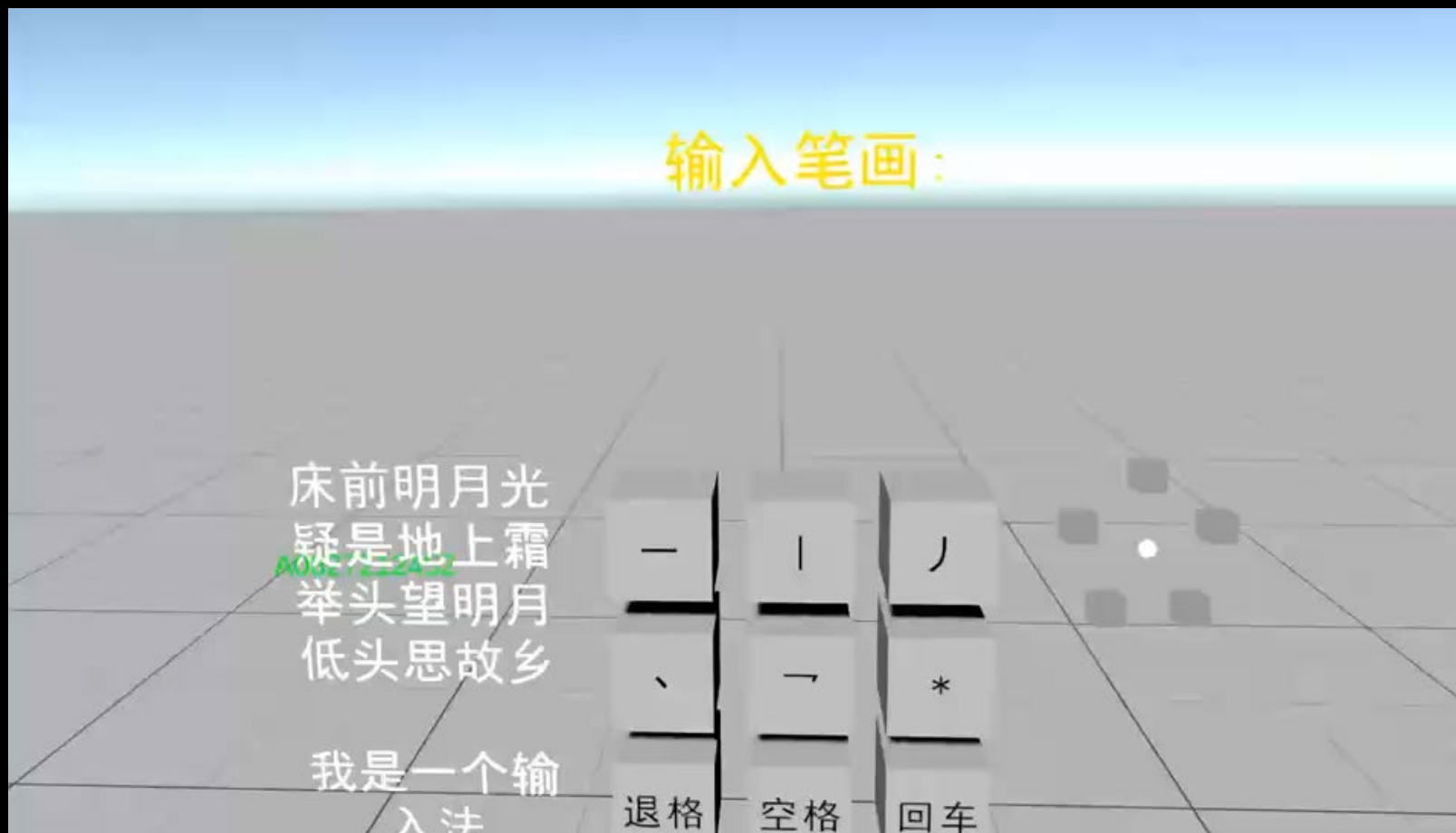
Prototype



Intended interaction

- 1 Choose a stroke
- 2 Inspect the composition
- 3 Select a candidate
- 4 Correct or commit

Prototype



Gesture and controller mappings

Core question

Which mapping makes composition and correction easier to learn and perform?

Future work

- Hand gesture recognition. Each finger is assigned with one stroke.
- Selects a candidate; a separate action confirms it.
- Connected to true IME
- Predictive phrase/word-level completion
- Accessibility

User Experiment Research

Stroke interaction: Compare mid-air drawing, controller gestures, and surface-based input for accuracy and ease of use.

Spatial ergonomics: Explore input position, scale, and arm support to reduce fatigue and maintain comfort.

Recognition flexibility: Investigate tolerance for variations in stroke order, direction, and personal writing habits.

Visual feedback: Test stroke trails, recognition previews, and depth cues for clarity without obstructing the environment.

Candidate selection and correction: Explore gaze, pinch, and gesture interactions for choosing characters and fixing errors.

Learnability: Study how users discover, remember, and become fluent with the stroke input method.

Attention and context: Evaluate input while users interact with virtual content or monitor their physical surroundings.

Evaluation: Measure entry speed, recognition accuracy, correction effort, workload, and fatigue over repeated sessions.

Multimodel Correction

Gaze + gesture: Look at an incorrect character and pinch to select or replace it.

Voice + stroke: Use spoken cues to clarify an ambiguous character or stroke.

Targeted stroke repair: Redraw only the incorrect stroke while preserving the rest of the character.

Cross-modal alternatives: Switch between voice, gaze, and hand input when one method fails.

Correction feedback: Preview the proposed change and provide simple confirmation and undo.

Research focus: Compare correction speed, accuracy, mental workload, and physical effort across modality combinations.

Security and Privacy

94.33%

55,541 users

Head and hand motion supported identification across sessions in a VR game study.

IME question: do text-entry movements also carry stable, person-specific information?

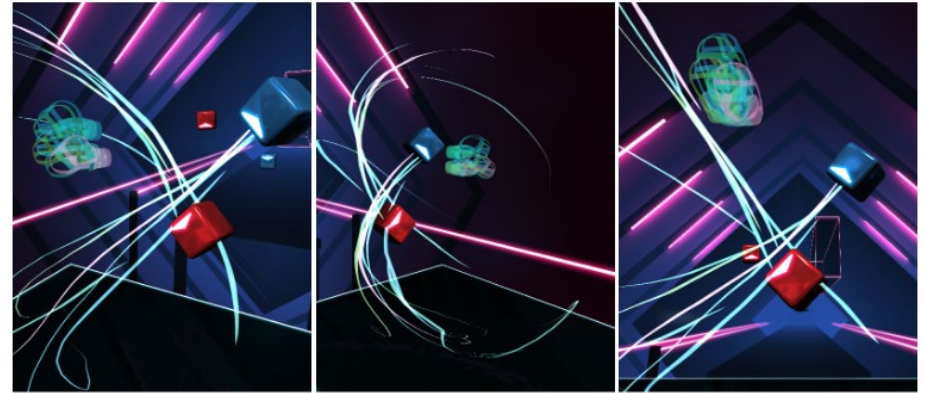


Figure 3: Five Beat Saber users hitting the same block pattern.

Security and Privacy

Part of the threat model	Proposed starting boundary
Observer	Another user in the shared virtual room
Visible data	Rendered hand movement
Private data	Candidate list and IME event log
Inference target	Entered stroke sequence, then text

Visible movement may expose entered text

Security and Privacy

Identification

Which person is this?

Authentication

Is this the person they claim to be?

Authorization

What may this person access or do?

Thanks for listening

- Q/A